

Taxi Trip Data — Analysis Summary

Cleaning, Visualization & Insights from the Seaborn "taxis" Dataset

Dataset & Cleaning

The dataset contains 6,433 taxi trips across 14 fields, spanning late February to March 2019. All numeric fields (fare, distance, tip, tolls, total, passengers) were complete with no missing values and no duplicate rows. Missing values were confined to categorical fields: **payment** (44 rows, ~0.7%), **pickup_zone** (26), **dropoff_zone** (45), **pickup_borough** (26), and **dropoff_borough** (45). All five were imputed using the **mode** (most frequent category) of their respective column, preserving the full 6,433-row dataset.

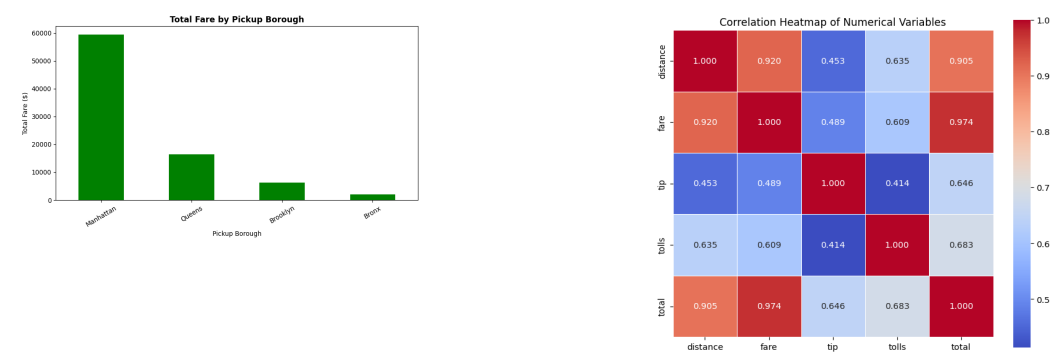
Key Findings

- **Manhattan dominates** taxi activity: 5,294 of 6,433 pickups (82%) originate there, generating \$59,426 in total fares — over 3.6x more than Queens, the next-highest borough (\$16,382).
- **Distance and fare are strongly correlated** ($r = 0.92$), and fare correlates even more strongly with the total charge ($r = 0.97$). Tip amount correlates more moderately with fare ($r = 0.49$), indicating tipping behavior is more variable than the fare-distance relationship.
- **Credit card is the dominant payment method**, used in 71.8% of trips versus 28.2% paid in cash.
- **Trip distance is right-skewed**: the median trip is 1.64 miles, well below the mean of 3.02 miles, with a long tail of trips extending up to 36.7 miles.
- **Tip amounts vary by borough** — the box plot shows wider spread and more high-value outliers for Manhattan pickups, consistent with its much larger trip volume.
- **Average fare fluctuates day-to-day** across the sampled month with no strong sustained trend, as shown in the fare-over-time line chart.

Visualizations Produced

Basic (Matplotlib)	Statistical (Seaborn)	Advanced (Seaborn)
Line — fare over time	Box — tip by borough	Heatmap — correlations
Bar — total fare by borough	Count — trips by borough	Pair plot — by pickup zone
Pie — payment method share	Scatter — distance vs fare	Violin — fare by payment
Histogram — distance distribution		

Supporting Charts



Left: total fare by pickup borough. Right: correlation heatmap of numerical variables.

Conclusion

Manhattan drives the overwhelming majority of taxi demand and revenue in this dataset, fares scale predictably with distance, and credit card is the clearly preferred payment method. These patterns suggest operational decisions — driver allocation, dynamic pricing, surge zones — should prioritize high-density Manhattan activity while still accounting for the smaller but higher-fare, longer-distance trips that occur elsewhere.